# Plain-English Guide to `factor_data.csv`

## What this dataset is

This dataset is a monthly panel of company-level stock and financial data. Each row appears to represent one company in one month, with columns for stock returns, market value, accounting measures, and pre-computed labels for common finance factors such as size, value, profitability, investment, and momentum.

## Column-by-column explanation

| Column | Meaning in simple terms |
|---|---|
| `co_code` | Company identifier. This is the ID used to track the same company across months. |
| `Month` | The calendar month for the row, in `YYYY-MM` format. |
| `monthly_gross_return` | The stock's gross monthly return. Values above 1 mean the stock gained value that month; values below 1 mean it lost value. |
| `mktcap` | Market capitalization, or the company's total market value. Usually calculated as share price times number of shares. |
| `price` | Share price for that month. |
| `equity_bv_on_stkdate` | Book value of equity measured on the stock date. Book value is an accounting-based estimate of company value. |
| `fy_book_value` | Fiscal-year book value. This is the book value taken from the company's financial statements for the fiscal year. |
| `lag_mv` | Lagged market value, meaning the market cap from the previous month or reference period. |
| `BM_sep` | Book-to-market ratio, a classic value signal. Higher values usually mean the stock is cheaper relative to its accounting value. |
| `Size_Label` | Size category based on market capitalization. In this dataset, `B` means big and `S` means small. |
| `BM_Label` | Book-to-market category. Here `G` means growth, `N` means neutral, and `V` means value. |
| `Year` | Calendar year for the observation. |
| `sa_net_worth` | Net worth from the company's financial accounts. A rough measure of how much the firm is worth after liabilities. |
| `sa_total_assets` | Total assets reported by the company. This includes everything the company owns or controls. |

| Column | Meaning in simple terms |
|---|---|
| `sa_sales` | Sales revenue. This is how much the company sold during the period. |
| `sa_cost_of_goods_sold` | Cost of goods sold. These are direct costs tied to producing what the company sells. |
| `sa_total_interest_exp` | Total interest expense. This is the cost of borrowing money. |
| `sa_selling_distribution_exp` | Selling and distribution expenses. These are costs for marketing, shipping, and getting the product to customers. |
| `Month_annual` | A month value used in annual alignment or year-based grouping. It appears to be a helper field rather than a core financial measure. |
| `Corrected_Year` | An adjusted year field used to line up accounting data with the correct month. |
| `Corrected_Month` | An adjusted month field used together with `Corrected_Year` for temporal alignment. |
| `lagged_book_equity` | Book equity measured from an earlier period. Lagged values are often used to avoid look-ahead bias. |
| `lagged_assets` | Total assets from an earlier period. |
| `lagged2_assets` | Total assets from two periods earlier. |
| `OpProf` | Operating profitability measure. In simple terms, it asks how efficiently the firm turns sales and assets into profit. |
| `Inv` | Investment measure. This captures how aggressively the firm is expanding its assets. |
| `OpProf_Label` | Operating profitability category. The observed labels are `R` (robust), `N` (neutral), and `W` (weak). |
| `Inv_Label` | Investment category. The observed labels are `C` (conservative), `N` (neutral), and `A` (aggressive). |
| `lag_ret` | Lagged return, usually the previous month's return. |
| `log_ret` | Log return, a transformed return measure that is easier to aggregate over time. |
| `Momentum` | Momentum score. This measures whether recent performance has been strong or weak. |
| `Mom_Label` | Momentum category. The observed labels are `L` (loser), `N` (neutral), and `W` (winner). |

## Short interpretation

At a high level, this dataset looks like a stock-market analysis dataset built for factor research or return prediction. It combines *what the stock did* (returns and prices) with *what the company looks like fundamentally* (assets, sales, book value, expenses) and then adds factor-style labels that group firms into categories like small vs. big, value vs. growth, robust vs. weak profitability, conservative vs. aggressive investment, and winner vs. loser momentum.

For CS students, the main idea is that each row is a monthly snapshot of a company, and the dataset is structured so that you can try tasks like clustering, classification, or forecasting without needing deep finance knowledge. The labels turn messy financial numbers into simpler buckets that are easier to model.

## Problem formulation

Let $i$ index a company and let $t$ index a month. For each company-month pair, we build an input window from the past $L$ months of factor information:

$$\mathbf{X}_{i,t} = \left[\mathbf{f}_{i,t-L+1}, \mathbf{f}_{i,t-L+2}, \ldots, \mathbf{f}_{i,t}\right],$$

where $\mathbf{f}_{i,t}$ collects the available predictors at month $t$, such as market capitalization, price, book-to-market, lagged accounting variables, profitability, investment, momentum, and the categorical factor labels.

The main prediction target is the one-month-ahead return:

$$y_{i,t+1} = r_{i,t+1} = \texttt{monthly\_gross\_return}_{i,t+1} - 1.$$

If you prefer to keep the target in gross-return form, then the same task can be written as predicting $g_{i,t+1} = \texttt{monthly\_gross\_return}_{i,t+1}$. In either case, the learning problem is to fit a function

$$\hat{y}_{i,t+1} = f_\theta(\mathbf{X}_{i,t})$$

that maps historical factor windows to future return forecasts.

Because the project is cross-sectional, the relative ranking of stocks often matters more than the exact numeric forecast. A common training objective is therefore regression loss such as mean squared error,

$$\theta^* = \arg\min_\theta \frac{1}{N} \sum_{i,t} \left(y_{i,t+1} - \hat{y}_{i,t+1}\right)^2,$$

or a ranking-oriented objective if the goal is portfolio construction rather than point prediction.

The factor buckets can also be used for ablation studies. For example, define a small-cap subset

$$\mathcal{S}_t = \{i : \texttt{Size\_Label}_{i,t} = \texttt{S}\},$$

and a large-cap subset

$$\mathcal{B}_t = \{i : \texttt{Size\_Label}_{i,t} = \texttt{B}\}.$$

You can then evaluate whether predictive signal is stronger inside one segment than another, and repeat the same idea for value versus growth, robust versus weak profitability, conservative versus aggressive investment, and winner versus loser momentum.